AgentOS — Full Product Specification

> *The Operating System for AI Agents*
> *Powered by: Agent Beacon + Everything Claude Code*

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Product Vision

> ***AgentOS*** *is the Operating System for AI Agents.*
> *Monitor Control Understand Evolve*

The 5 Pillars

Layer	Description
Observe	See everything happening in your agent system
Control	Manage agents, commands, hooks, skills, MCPs
Understand	ROI, model ranking, quality scoring
Evolve	Self-healing, auto-discovery, model routing
Collaborate	Team mode, marketplace, blueprints

Data Sources

```
Agent Beacon       Telemetry, session data, token usage
Everything CC      Agent configs, commands, hooks, skills
AgentOS Engine     Intelligence layer connecting both
```

Architecture Overview

```
                    AgentOS Platform


    OBSERVE — See everything
Dashboard, Tokens, Graph, Sessions, Deps


    CONTROL — Manage everything
Agents, Commands, Hooks, Skills, Budget, Versions


    UNDERSTAND — Analyze everything
ROI, Models, Quality, Alerts, Failures


    EVOLVE — Improve automatically
```

Auto-Discovery, Router, Self-Healing, What-If

COLLABORATE — Work together
Team, Marketplace, Plugins, Shared Configs

Powered by: Agent Beacon + Everything Claude Code

Self-Evolving Feedback Loop

Auto-Discover Chat-to-Config Visual Graph Model Router

Self-Healing Quality Judge ROI Analytics Budget Control

Continuous Feedback Loop

Project Structure

```
agentOS/
    apps/
        web/                    # Next.js 14 Dashboard
            app/
                (dashboard)/
                    overview/
                    agents/
                    commands/
                    hooks/
                    skills/
                    tokens/
                    sessions/
                    graph/
                    budget/
                    models/
                    roi/
                    settings/
```

```
                chat/
            layout.tsx
          components/
              ui/             # shadcn components
              charts/         # Recharts/D3
              graph/          # React Flow
              shared/
          lib/
           api.ts
           websocket.ts
           telemetry.ts

      api/                  # Go Backend
         handlers/
         services/
         models/
         beacon/            # Beacon integration
         config-engine/     # Claude Code config
         websocket/

   packages/
      shared-types/          # TypeScript types
      config-parser/         # YAML/JSON parser
      telemetry-sdk/         # OTel wrapper

   docs/
```

Sprint 0 — Foundation & Architecture

Duration: 1 week

Story 0.1 — Project Setup & Monorepo

As a developer,
I want the project structure ready,
So that I can develop features quickly.

Tasks:

- ☐ Setup Next.js 14+ with App Router

- [ ] Setup Tailwind + shadcn/ui
- [ ] Setup Go backend (compatible with Beacon)
- [ ] Setup SQLite local-first + PostgreSQL config
- [ ] Setup WebSocket infrastructure
- [ ] Setup OpenTelemetry SDK
- [ ] Connect to Agent Beacon data pipeline
- [ ] Connect to Everything Claude Code config files

Story 0.2 — App Shell & Navigation

As a user,
I want a main layout with navigation,
So that I can move between sections.

Prototype:

```
 AgentOS                                  User


  HOME
Overview

                      Welcome to AgentOS
OBSERVE

  Tokens          "The Operating System for
  Graph            your AI Agents"
  Sessns
  Deps           [    Quick Setup ]
                [    Connect Beacon ]
                  [    Import Config ]
CONTROL

  Agents
  Cmds
  Hooks
  Skills
  Chat
  Budget
```

Versns

UNDERSTAND

ROI
Models
Qualty
Alerts

EVOLVE

Discvr
Router
Heal
WhatIf

COLLABORATE

Team
Market
Plugns

Sprint 1 — Overview Dashboard

Duration: 2 weeks

Story 1.1 — Summary Cards

As a user,
I want to see system status at a glance,
So that I know everything is working.

Acceptance Criteria:

- Show count of active agents

- Show today's sessions count
- Show total token usage + cost today
- Show System Health Score
- Show currently active model

Prototype:

```
 HOME        Dashboard Overview
Overview


          Agents          Sessions          Cost            Hlth
                                    Today
        7 active        23 today        $4.82          94%
        2 idle          3 live             12%


      Active Model: Claude Sonnet 4    [Change   ]


        Token Usage — Last 7 Days

      5K
      4K
      3K
      2K
      1K

         Mon Tue Wed Thu Fri Sat Sun
       Input     Output     Cache
```

Story 1.2 — Recent Sessions Feed

As a user,
I want to see recent sessions,

So that I know what agents have been doing.

Prototype:

```
 Recent Sessions                [See All]

 #2891  Refactor auth module
   code-writer     4m 23s    $0.34     Pass
12 min ago

 #2890  Fix login bug
   debugger     2m 11s    $0.18     Pass
34 min ago

 #2889  Add API endpoint
   code-writer     8m 45s    $0.72     Fail
1 hour ago
```

Story 1.3 — Environment Profile Card

As a user,
I want to see my auto-detected environment profile,
So that I know what the system has discovered.

Prototype:

```
  Environment Profile       [   Re-scan]

Stack:     Next.js 14 + TypeScript
Backend:   Node.js + Prisma + PostgreSQL
CI/CD:     GitHub Actions
Testing:   Jest (coverage: 62%)
Linting:   ESLint + Prettier
Docker:    Yes (docker-compose)
Last Scan: 2 hours ago

   3 Recommendations Available
```

Missing: Security Scanner Hook
Missing: Test Coverage Agent
Upgrade: Prisma MCP (v1.2 v1.4)

[View Details] [Apply All]

Story 1.4 — Active Alerts Widget

As a user,
I want to see active alerts on the dashboard,
So that I can quickly spot problems.

Prototype:

Active Alerts (3) [See All]

CRITICAL Agent "researcher" in loop
15 iterations, $0.89 spent
[Kill] [View Session]

WARNING Budget 80% used (Backend team)
$160/$200 this month
[Details] [Increase]

WARNING GPT-4o API key expires in 3 days
[Renew] [Dismiss]

Sprint 2 — Token Analytics

Duration: 2 weeks

Story 2.1 — Token Overview Page

As a user,
I want to see overall token usage,

So that I can manage my costs.

Prototype:

```
Tokens      Token Analytics

      Period: [Today   ] [This Week] [This Month] [Custom]


  Total Tkns     Total Cost     Avg/Sess     Efficiency
  847,231        $12.84         3,421        87%
   8%            3%             12%          5%



    Usage Over Time

  50K
  40K
  30K
  20K

     1  3  5  7  9  11 13 15 17
     Input      Output  ·· Forecast
```

Story 2.2 — Token Usage by Agent

As a user,
I want to see how many tokens each agent used,
So that I can identify expensive agents.

Prototype:

```
      Usage by Agent

  Agent        Tokens    Cost   Sessions
```

```
code-writer 525,342   $7.96   89
researcher  152,501   $2.31   34
reviewer    101,677   $1.54   56
tester       67,711   $1.03   42

Bar:   62%   18%   12%   8%
```

Story 2.3 — Token Usage by Model

As a user,
I want to compare token usage across models,
So that I can make cost-efficient model choices.

Prototype:

```
Usage by Model

Claude Sonnet  534K  $8.01  63%


Claude Opus    201K  $6.03  24%


GPT-4o         112K  $2.24  13%

  API Key expired — reconnect?

Switching code-writer to Sonnet could
save ~$2.40/month with similar quality
```

Story 2.4 — Single Session Token Detail

As a user,
I want to see token breakdown for a specific session,

So that I know where tokens went.

Prototype:

```
   Session #2891 Token Breakdown
Total: 14,231 tokens ($0.34)

Step              In      Out     Cost

1. Read files        1,204     342   $0.02
2. Plan              2,891   1,456   $0.06
3. Edit auth.ts        892   2,103   $0.05
4. Edit middleware     445   1,234   $0.03
5. Run tests           334     567   $0.01
6. Fix test failure  1,023   1,402   $0.04
7. Verify            1,892     446   $0.03

[Read][Plan][    Edit    ][Mid][Tst][Fix][Verify]
```

Sprint 3 — Agent Manager

Duration: 2 weeks

Story 3.1 — Agent List Page

As a user,
I want to see all my agents in one place,
So that I know what agents I have.

Prototype:

```
Agents       Agent Registry
        [+ New Agent]  [   Import]     Search...
        Filter: [All   ] [Active] [Idle] [Error]
```

```
  code-writer                    Active
Writes and edits code based on plans
Model: Sonnet    Sessions: 89    Cost: $7.96
Success Rate: 94%       Quality: 8.4/10
[   Config] [   Stats] [   Pause] [   Delete]

  researcher                     Active
Researches codebase and docs for context
Model: Sonnet    Sessions: 34    Cost: $2.31
Success Rate: 88%       Quality: 7.9/10
[   Config] [   Stats] [   Pause] [   Delete]

  reviewer                       Idle
Reviews PRs and code changes
Model: Opus     Sessions: 56    Cost: $1.54
Success Rate: 96%       Quality: 9.1/10
[   Config] [   Stats] [   Pause] [   Delete]
```

Story 3.2 — Agent Detail / Config Page

As a user,
I want to see and edit full agent details,
So that I can configure agent behavior.

Prototype:

```
Agents      code-writer                    Active
 Detail    [General] [Skills] [Hooks] [MCPs] [Stats] [Logs]

     General Settings
Name:       [code-writer              ]
Description: [Writes and edits code based  ]
             [on plans and specifications  ]
Model:      [Claude Sonnet 4          ]
Max Tokens:  [100000                  ]
Temperature:              0.3
```

System Prompt:

You are a senior developer who writes
clean, tested, production-ready code...

Rules:
Follow TDD approach
Max 200 lines per file
Always add JSDoc comments
[+ Add Rule]

[Save Changes] [Reset] [Export YAML]

Story 3.3 — Create New Agent Wizard

As a user,
I want to create a new agent,
So that I can automate a specific task.

Prototype:

Agents Create New Agent
New Step 2 of 4: Define Purpose

Name Purpose Config Review

Describe It

I want an agent that reviews my database
migrations before they run. It should
check for destructive changes and missing
indexes, and suggest rollback plans.

```
I'll create:
Agent: "migration-reviewer"
Hook: pre-migration trigger
Skill: SQL analysis
MCP: Database connector (suggested)

  [   Back]  [Preview Config   ]
```

Sprint 4 — Command Center

Duration: 2 weeks

Story 4.1 — Command List Page

As a user,
I want to see all my commands,
So that I know what commands are available.

Prototype:

```
Cmds        Command Center
      [+ New Command]    Search commands...


    /plan                      Built-in
  Creates a detailed implementation plan
  Usage: /plan <description>
  Used: 45 times    Avg tokens: 2,340
  [   Docs] [   Edit] [   Stats]

    /tdd                       Built-in
  Test-driven development implementation
  Usage: /tdd <feature>
  Used: 32 times    Avg tokens: 4,120
  [   Docs] [   Edit] [   Stats]
```

```
  /verify                    Built-in
Verifies implementation against the plan
Usage: /verify
Used: 28 times   Avg tokens: 1,890
[  Docs] [  Edit] [  Stats]

  /review-pr                 Custom
Reviews a pull request for code quality
Usage: /review-pr <PR-number>
Used: 12 times   Avg tokens: 3,210
[  Docs] [  Edit] [  Stats] [  Delete]
```

Story 4.2 — Command Detail & Documentation

As a user,
I want to see full docs for each command,
So that I know how to use it.

Prototype:

```
    Documentation

  What does /plan do?
Creates a comprehensive implementation plan before
any code is written. Analyzes codebase and creates
a step-by-step action plan saved to PLAN.md

  Usage: /plan <what you want to build>

  Examples:
/plan Add user authentication with JWT
/plan Refactor payment module to strategy pattern
/plan Add rate limiting to all API endpoints

  Steps:
1. Scan project structure
```

2. Identify relevant files
3. Analyze dependencies
4. Create ordered task list
5. Estimate effort per task
6. Save plan to PLAN.md

Related: /tdd /verify

Story 4.3 — Create New Command

As a user,
I want to create a new command,
So that I can automate my specific workflows.

Prototype:

Describe your command

I want a command that generates API docs from my code comments. Scan all route files, extract JSDoc comments, create a markdown docs file.

Generated Config:

```
name: /gen-docs
description: Generate API docs from comments
agent: code-writer
steps:
  - scan: "src/routes/**/*.ts"
  - extract: jsdoc_comments
  - generate: "docs/API.md"
```

[Edit Config] [Test Run] [Create]

Sprint 5 — Hook Manager

Duration: 1.5 weeks

Story 5.1 — Hook List & Management

As a user,
I want to manage all my hooks,
So that I can control my automation workflow.

Prototype:

```
Hooks        Hook Manager
        [+ New Hook]  Filter: [All    ] [Pre] [Post] [Active]


          session-start                Pre-session
        Loads context from previous sessions
        Runs: Every session     Avg: 1.2s
        Status:     Active     Failures: 0
        [    Config] [    Logs] [    Disable]

          pre-commit-check              Pre-commit
        Runs linter and type-check before commit
        Runs: 156 times     Avg: 3.4s
        Status:     Active     Failures: 12 (7.7%)
        [    Config] [    Logs] [    Disable]

          session-end                  Post-session
        Saves context and generates summary
        Runs: Every session     Avg: 0.8s
        Status:     Active     Failures: 2
        [    Config] [    Logs] [    Disable]
```

Story 5.2 — Hook Detail & Config Editor

As a user,
I want to edit hook configuration details,
So that I can fine-tune automation triggers.

Prototype:

```
    Config
Name:    [pre-commit-check          ]
Trigger:  [Before Git Commit         ]

Condition:

  files.changed.some(f =>
   f.endsWith('.ts') || f.endsWith('.tsx'))


Actions:
1   Run: eslint --fix {changed_files}
2   Run: tsc --noEmit
3   Run: jest --related {changed_files}
[+ Add Action]

On Failure: [   Block commit    Warn only]
Timeout:   [30] seconds

[   Save] [   Test Run] [   Reset]
```

Sprint 6 — Skill & MCP Store

Duration: 2 weeks

Story 6.1 — Browse Page

As a user,
I want to browse available Skills and MCPs,
So that I can extend my agents' capabilities.

Prototype:

```
Skills       Skill & MCP Store
          [Installed (8)] [Available (47)] [Updates (3)]
            Search...  Category: [All   ] [Database] [Git]...

              Installed

              GitLab MCP                PostgreSQL MCP
                    (4.8)                     (4.2)
             v2.1.0    Installed       v1.3.0    Installed
             Used by: 3 agents         Used by: 2 agents
             [   Config] [   Upd]      [   Config] [   Rm]


              Available

              Jira MCP                  Security Scanner
                    (4.3)                     (4.7)
              2,341 installs            5,892 installs
             Compatible                Compatible
             [   Install] [  ]         [   Install] [  ]
```

Story 6.2 — Detail Page with Impact Analysis

As a user,
I want to see impact analysis before installing/removing an MCP,
So that I understand what will change.

Prototype:

```
          Overview
        Security Scanner Skill
    Scans for SQL injection, XSS, hardcoded secrets...

        Compatible With:
        Your Stack: Next.js + TypeScript
        Your Agents: code-writer, reviewer
```

Impact Analysis (if installed):

code-writer will auto-check security
reviewer will include security report
pre-commit hook will add ~2s
Est. additional cost: $1.20/month

[Install Now] [Add to Queue]

Sprint 7 — Visual Flow Graph

Duration: 2 weeks

Story 7.1 — Agent Pipeline Graph View

As a user,
I want to see a visual graph of agents and hooks,
So that I understand the execution flow.

Prototype:

Graph Agent Flow Graph
[View] [Edit Mode] [Live Mode] [Zoom: 100%]

User
Prompt

Orchestratr 2.1K

```
                Plannr      Rsch      Tstr
                  1.4K        3K      0.8K



              pre-edit


              Writer              5.2K


              post-edt


              Verifyr             2.8K


        Token Heat:  Low  Med  High
          Bottleneck:  Researcher (3.2s avg)
```

Story 7.2 — Drag & Drop Graph Editor

As a user,
I want to edit the pipeline visually using drag & drop,
So that I don't need to write YAML manually.

Prototype:

```
      Palette              Canvas

       Agent
    drag  add             Planner
```

```
  Hook
drag  add                    pre-edit

  Conditn
drag  add
                  New Agent
  Loop             Name: [_____]
drag  add          Type: [Code Writer ]
                 Model:[Sonnet      ]


          [  Save Pipeline] [  Test]
          [  Export YAML]  [  Undo]
```

Sprint 8 — Chat-to-Config

Duration: 2 weeks

Story 8.1 — Chat Interface for Configuration

As a user,
I want to configure my system using natural language,
So that I don't need to fill out forms.

Prototype:

```
Chat       AgentOS Assistant


          Hi! I can create agents, hooks, commands,
          install skills/MCPs, and debug issues.
        What would you like to do?

                          PR
                                      merge
```

! :

1. Agent: security-reviewer
   Model: Claude Opus
   [Preview] [Edit] [Accept]

2. Hook: pre-merge
   [Preview] [Edit] [Accept]

3. MCP: GitHub/GitLab Integration
   [Preview] [Edit] [Accept]

4. Skill: Security Scanner
   [Preview] [Edit] [Accept]

Est. cost: ~$3.20/month
[Apply All (4)] [Export Blueprint]

Type your request... [Send]

Story 8.2 — Conversational Debugging

As a user,
I want to debug agent issues through chat,
So that I don't have to manually investigate logs.

Prototype:

. :

Issue 1: Hook "pre-edit" is slow

Avg time: 4.2s (was 1.1s last week)
Cause: Linting entire project vs changed files
Fix: Scope lint to changed files only
[Auto-Fix] [Show Diff]

Issue 2: "researcher" reads 340 files/sess
(up from 45 after /docs folder added)
Fix: Exclude /docs from search scope
[Auto-Fix] [Show Details]

Issue 3: Cache hit rate 12% (should be 60%)
Fix: Enable session-end hook for caching
[Auto-Fix] [Show Details]

[Fix All 3 Issues]

Sprint 9 — Budget & Safety

Duration: 2 weeks

Story 9.1 — Budget Dashboard

As a user,
I want to manage budgets by agent and team,
So that I can control costs.

Prototype:

Budget Budget & Safety Control
[Budget] [Kill Switches] [Alerts Config]

Monthly Budget Overview

```
Global:  $142 / $500        [                ] 28%
  Backend Team:  $89 / $200  [                ] 45%
    code-writer:  $52        [                ] 87%
    researcher:   $23        [                ] 46%
    tester:       $14        [                ] 28%
  Frontend Team: $38 / $150  [                ] 25%
  DevOps Team:   $15 / $150  [                ] 10%

Forecast: ~$340 by month end (within budget)
code-writer may exceed limit in 4 days
```

Story 9.2 — Kill Switch Configuration

As a user,
I want to set auto-stop rules for agents,
So that agents can't go out of control.

Prototype:

```
     Kill Switch Rules
[+ Add Rule]

IF [session_cost     ] [>    ] [$1.00   ]
THEN [   Kill Session   ]
AND  [   Send Alert    ]
Triggered: 3 times    Last: 2 days ago

IF [loop_iterations   ] [>    ] [5       ]
THEN [   Kill Session   ]
AND  [   Send Alert    ]
Triggered: 1 time     Last: 5 days ago

IF [daily_spend       ] [>    ] [80%     ]
THEN [   Downgrade Model   ] Opus    Sonnet
Triggered: 0 times
```

```
EMERGENCY: STOP ALL AGENTS
[  EMERGENCY STOP]
```

Story 9.3 — Cost Prediction Before Execution

As a user,
I want to see cost estimate before a task runs,
So that I can approve or cancel expensive tasks.

Prototype:

```
  Pre-Execution Cost Estimate
 Task: "Refactor payment module"
 Agent: code-writer      Model: Claude Sonnet

Input Tokens:  ~12,000   $0.04
Output Tokens:  ~8,500   $0.13
Tool Calls:    ~5        $0.02

Total Estimate:          $0.19
Confidence: 78%   Range: $0.12 - $0.34

  Alternative: Use Haiku for $0.03
  (quality may drop ~15% for this task type)

Budget remaining today: $4.82
[  Execute ($0.19)] [  Use Cheaper] [  Cancel]
```

Sprint 10 — Session Explorer & Time-Travel

Duration: 2.5 weeks

Story 10.1 — Session List & Explorer

As a user,
I want to browse and filter all sessions,
So that I can find and analyze specific runs.

Prototype:

```
Sessns       Session Explorer
          Search...    Agent [All   ]    Status [All   ]

        #2891  Refactor auth      writer       Pass
               4m 23s    $0.34     Sonnet     8.5
             Timeline:
             [   View] [   Replay] [   Branch]

        #2890  Fix login bug      debugger     Pass
               2m 11s    $0.18     Sonnet     9.1
             Timeline:
             [   View] [   Replay] [   Branch]

        #2889  Add API endpoint   writer       Fail
               8m 45s    $0.72     Opus       4.2
             Timeline:
               Loop detected at step 5
             [   View] [   Replay] [   Branch]
```

Story 10.2 — Time-Travel Session Debugger

As a user,
I want to step through a session's history,
So that I can find where an agent went wrong.

Prototype:

```
Sessns       Session #2889 — Time-Travel Debugger
 Detail    Task: "Add API endpoint for user preferences"
         Agent: code-writer    Result:    Failed
```

```
Timeline:
        T1  T2  T3  T4 [T5] T6  T7    selected

        Step T5: Edit routes.ts
       45s        $0.12      Tokens: 2,341 / 1,892

        Agent Thought:
      "I need to add a new route for user prefs.
       Looking at the existing pattern, I should
       also modify the auth middleware..."

        Tool Calls:
            read_file("src/routes/index.ts")
            read_file("src/middleware/auth.ts")
            edit_file("src/routes/preferences.ts")

        Issue: Missing input validation
       This led to test failure in T7


        [   Branch from T5]
       Override: Add input validation using zod
       [   Run New Branch]  [   Est: $0.08]

    [   Prev] [   Next] [   Revert to T4]
```

Story 10.3 — Branch Comparison View

As a user,
I want to compare two execution paths,
So that I can learn which approach is better.

Prototype:

```
        Branch Comparison

   Original (   Failed)       Branch A (   Passed)
```

```
                                                  '     '     '

  Steps:    7              Steps:    7
  Tokens:   14.2K          Tokens:   11.8K
  Cost:     $0.72          Cost:     $0.41
  Quality:  4.2            Quality:  8.7
  Result:                  Result:

  Key Diff at T5:
  Original: No input validation
  Branch A: Added zod schema

     Add rule "always validate input" to code-writer?
  [   Add Rule Automatically]
```

Sprint 11 — Model Intelligence

Duration: 2 weeks

Story 11.1 — Model Ranking Dashboard

As a user,
I want to see model rankings,
So that I can choose the best model.

Prototype:

```
  Models       Model Intelligence
          [Ranking] [Comparison] [Router Config] [Benchmark]

          #  Model          Status  Score  Avg Cost

          1  Claude Sonnet 4         94/100  $0.015/K

            Currently active for: 4 agents

          2  Claude Opus 4           91/100  $0.075/K
```

```
  Currently active for: 1 agent

3  GPT-4o              82/100  $0.025/K

   API key expired 2 days ago  [   Reconnect]

4  DeepSeek R1         76/100  $0.004/K


5  Llama 4             ??/100  $0.002/K
   Benchmarking... 34% complete
 [   Pause] [View Progress]

[+ Add Model]  [   Re-benchmark All]
```

Story 11.2 — Ranking by Task Type

As a user,
I want to know which model excels at which task type,
So that I can route tasks intelligently.

Prototype:

```
Task Type        Best        Runner-up     Third

  Code Generation  Opus (96)   Sonnet (91)   GPT-4o (84)
  Code Review      Opus (94)   GPT-4o (88)   Sonnet (85)
  Bug Fixing       Sonnet (93) Opus (90)     DeepSk (72)
  Documentation    Sonnet (92) GPT-4o (87)   DeepSk (81)
  Test Writing     Sonnet (95) Opus (89)     GPT-4o (82)
  Refactoring      Opus (97)   Sonnet (88)   GPT-4o (79)

  You use Opus for Bug Fixing but Sonnet scores higher
  AND costs 80% less for this task type.
[   Switch Automatically]
```

Story 11.3 — Auto-Model Router Configuration

As a user,
I want models to be auto-selected based on task type,
So that I always get the best quality/cost balance.

Prototype:

```
    Auto-Model Router
Mode: [  Automatic   Manual   Budget-First]

Task Type       Primary   Fallback

Code Generation  Opus      Sonnet
Code Review      Opus      GPT-4o
Bug Fixing       Sonnet    Opus
Documentation    Sonnet    DeepSeek
Test Writing     Sonnet    Opus
Refactoring      Opus      Sonnet
Simple Tasks     DeepSeek  Sonnet

Budget-Aware: When budget > 80%:
[  Downgrade all to cheapest capable model]
[  Only downgrade non-critical tasks]
[  Ask me every time]

[  Save Router Config]
```

Sprint 12 — ROI Analytics & Quality

Duration: 2 weeks

Story 12.1 — ROI Dashboard

As a user/manager,
I want to see the return on investment of my agents,
So that I can justify the costs.

Prototype:

ROI ROI Analytics Period: [This Month]

Spent	Saved	Value	ROI
$142	87 hrs	$4,350	2,963%

ROI Trend (6 Months)

4000%
3000%
2000%
1000%
500%

Jun Jul Aug Sep Oct Nov

Bugs Caught Early: 34
Est. post-production fix cost: $6,800
You saved: $6,658 this month

Monthly Narrative Report
"This month, agents saved 87 hours. Biggest win: auth refactor in 45min vs 2 days. 34 bugs caught before production."
[Export PDF] [Email to Team]

Story 12.2 — Agent Quality Scorecard

As a user,
I want to see quality metrics for each agent,
So that I know which agents need improvement.

Prototype:

```
  code-writer — Overall Quality:    8.4 / 10

Metric           Score    Trend

Code Correctness    9.1          0.3
Best Practices      8.7          0.1
Test Coverage       7.2          0.2
Security            8.9          same
Performance         8.1          0.5
Readability         8.5          same

Success Rate        94%          2%
User Overrides      12%          3%  (agent was wrong)
False Positives     7%           1%

  Judge Analysis:
"Test coverage dropping — new endpoints not getting
 integration tests. Suggest: always write integration
 test for new endpoints."
[   Apply Suggestion]  [   Full Report]
```

Sprint 13 — Auto-Discovery & Self-Healing

Duration: 2 weeks

Story 13.1 — Environment Auto-Discovery

As a user,
I want the system to auto-scan my environment,
So that I get relevant config recommendations.

Prototype:

```
Discvr       Auto-Discovery
        [Environment] [Drift Detection] [Recommendations]

        Last scan: 14 min ago  [   Re-scan]
```

```
Auto-scan: Every 6 hours

  Detected Stack:
Framework:  Next.js 14.2.1
Language:   TypeScript 5.4
ORM:        Prisma 5.10
Database:   PostgreSQL 16
Cache:      Redis 7.2          NEW
Auth:       NextAuth 5.0
Testing:    Jest + Playwright
CI/CD:      GitHub Actions

  Drift Detected! Redis added to project

  Recommendations:
    Install Redis MCP for cache management
    Add Redis monitoring to DevOps agent
    Create cache invalidation hook
[   Apply All]  [   Review Each]
```

Story 13.2 — Self-Healing Dashboard

As a user,
I want the system to detect and fix config issues automatically,
So that problems resolve with minimal manual work.

Prototype:

```
      Self-Healing Engine
Status:    Active    Issues Found: 3
Auto-fix: [   Suggest Only    Auto-apply]

      Auto-fixed
  Hook "pre-commit" timing out
  Root cause: lint scope too broad
  Fix: scoped to changed files     4.2s    0.8s
  [   Undo]
```

```
    Needs Approval
  "researcher" has 30% failure rate
  Cause: reading deprecated API docs
  Fix: exclude /docs/deprecated/
  [   Apply Fix] [   Dismiss] [   Details]


  This Month:
Detected: 12   Auto-fixed: 7   Approved: 3   Dismissed: 2
```

Sprint 14 — Dependency Map & Config Versioning

Duration: 1.5 weeks

Story 14.1 — Dependency Map

As a user,
I want to see how agents, skills, and hooks are connected,
So that I understand the impact of changes.

Prototype:

```
Deps        Dependency Map
       [Graph View   ] [Table View   ] [Impact Analysis]


              code-writer                    test-runner

                              GitLab MCP

              pre-edit
```

reviewer
code-analysis

Impact: Removing GitLab MCP affects
2 agents (code-writer, reviewer)

Story 14.2 — Config Version Control

As a user,
I want to see config change history and rollback,
So that I can undo bad configurations.

Prototype:

Current Version: v14 [Export] [Import]

v14 Today 2:34 PM Current
Added: security-reviewer agent + pre-merge hook
Source: Chat-to-Config
[View] [Diff with v13]

v13 Yesterday 5:12 PM
Modified: code-writer system prompt + pre-commit scope
Source: Self-Healing auto-fix
[View] [Rollback] [Diff]

v12 3 days ago
Added: tester agent + /tdd command
Source: Manual (Dashboard)
[View] [Rollback] [Diff]

Diff v13 v14
agents/security-reviewer.yaml Added
+ name: security-reviewer

```
+ model: claude-opus
hooks/pre-merge.yaml                    Added
+ trigger: before_merge
+ on_failure: block_merge
```

Sprint 15 — Team & Collaboration

Duration: 2 weeks

Story 15.1 — Team Dashboard

As a team admin,
I want to see team usage and performance,
So that I can manage costs and productivity.

Prototype:

```
Team        Team Management
        [Members] [Shared Config] [Permissions] [Analytics]

        Team: "Acme Engineering"    Members: 8
        Total Cost: $342/month      Active Agents: 12

        Member       Role    Cost    Sessions   Score

          Ali        Admin   $89     145        8.7
          Sara       Dev     $67     98         9.1
          Reza       Dev     $52     87         8.4
          Mina       Dev     $45     76         8.2
          Hassan     Lead    $38     45         9.0
          Zahra      Junior  $23     34         7.8
          Mohammad   DevOps  $18     23         8.5
          Fatemeh    QA      $10     12         9.3

        [+ Invite Member]  [   Team Settings]
```

Story 15.2 — Blueprint Marketplace

As a user,
I want to browse and install ready-made blueprints,
So that I can get started quickly with proven configurations.

Prototype:

```
Market       Blueprint Marketplace
        [Popular] [New] [Your Stack: Next.js] [Your Uploads]


          Full-Stack Next.js          Python ML Pipeline
                 (4.9)                       (4.4)
          5 agents, 8 hooks,          3 agents, 4 hooks
            1,234 installs            Not compatible
          Compatible                   (needs Python)
          [   Install] [  ]          [   Preview]


          Security Hardened           TDD Master
                 (4.8)                       (4.7)
            2,891 installs              987 installs
          Compatible                   Compatible
          [   Install] [  ]          [   Install] [  ]


        [   Create Blueprint from Current Config]
```

Sprint 16 — What-If Simulator & Alerts

Duration: 1.5 weeks

Story 16.1 — What-If Simulator

As a user,
I want to simulate changes before applying them,
So that I can predict impact without risk.

Prototype:

```
WhatIf       What-If Simulator


              "What if I switch code-writer Opus  Sonnet?"

          [   Simulate]

               Simulation Results
            Based on 234 historical sessions:

            Metric       Current(Opus) Predicted(Sonnet)

            Quality Score    9.2         8.7 (  5%)
            Success Rate     96%         92% (  4%)
            Avg Speed        45s         32s (  29%)
            Monthly Cost     $52         $11 (  79%)

            Quality by Task:
            Code Generation:  9.4    8.9 (minor)
            Refactoring:      9.6    8.1 (significant)
            Bug Fixing:       8.8    9.0 (better!)

              Switch everything except refactoring.
              Save $41/month, only 2% quality drop.

            [   Apply Recommendation] [   Full Switch]
```

Story 16.2 — Smart Alerts Configuration

As a user,
I want to configure intelligent alerts,
So that I'm notified when things go wrong.

Prototype:

```
      Alert Rules
[+ New Alert Rule]

   Token Spike Detection
When: Token usage > 200% of daily average
Alert: Email + Dashboard notification
Triggered: 2 times this month
[   Edit] [   Pause] [  ]

   Agent Loop Detection
When: Same tool called > 5 times in a loop
Action: Auto-kill + alert
Triggered: 1 time this month
[   Edit] [   Pause] [  ]

   Quality Drop Alert
When: Agent quality score drops > 10%
Alert: Dashboard + suggest investigation
Triggered: 0 times
[   Edit] [   Pause] [  ]

   Model Disconnect Alert
When: Model API errors > 3 times
Action: Auto-switch to fallback + alert
Triggered: 1 time (GPT-4o key expired)
[   Edit] [   Pause] [  ]
```

Sprint Summary

#	Sprint	Duration	Stories
0	Foundation & Architecture	1 week	2

1	Overview Dashboard	2 weeks	4
2	Token Analytics	2 weeks	4
3	Agent Manager	2 weeks	3
4	Command Center	2 weeks	3
5	Hook Manager	1.5 weeks	2
6	Skill & MCP Store	2 weeks	2
7	Visual Flow Graph	2 weeks	2
8	Chat-to-Config	2 weeks	2
9	Budget & Safety	2 weeks	3
10	Session Explorer & Time-Travel	2.5 weeks	3
11	Model Intelligence	2 weeks	3
12	ROI Analytics & Quality	2 weeks	2
13	Auto-Discovery & Self-Healing	2 weeks	2
14	Dependency Map & Versioning	1.5 weeks	2
15	Team & Marketplace	2 weeks	2
16	What-If Simulator & Alerts	1.5 weeks	2
—	**TOTAL**	**~30 weeks**	**41**

Tech Stack

Layer	Technology	Reason
Frontend	Next.js 14 + App Router	Modern, fast
UI	Tailwind + shadcn/ui	Beautiful, flexible
Charts	Recharts + D3.js	Token graphs
Graph	React Flow	Pipeline visualization

Backend	Go	Compatible with Beacon
Database	SQLite (local) + PostgreSQL (team)	Local-first
Realtime	WebSocket	Live monitoring
Telemetry	OpenTelemetry	Compatible with Beacon
Auth	NextAuth.js	Team mode
Config	YAML/JSON	Claude Code compatible

Key Design Principles

1. **Local-first** — Works offline, syncs when online
2. **Config as code** — Everything exportable as YAML/JSON
3. **Zero-touch** — Auto-discovers and configures itself
4. **Closed-loop** — Data Insight Action Better Data
5. **Non-intrusive** — Observes without changing agent behavior
6. **Progressive** — Works with basic setup, gets smarter over time

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AgentOS — The Operating System for AI Agents
Powered by Agent Beacon + Everything Claude Code